Evan McKinney
Iowa State Fall 2020

EE 494 Cumulative Reflection

During my undergraduate career in computer engineering, physics, and math at Iowa State University, I have been involved with multiple honor societies, undergraduate research, and leadership positions, always with the long-term goal of graduate studies in my mind. I can only envision my future being in academia as a professor and have used that to motivate myself when pushing through challenging semesters. My intention is to study and design quantum computer architectures as a PhD student.

It was around junior high when I learned I wanted to become a scientist. I remember watching a documentary on space during school and dreamed of life as an expert in cosmology being interviewed for an educational tv program. Unfortunately, I was not a particularly motivated student for most of high school. I joined the debate team on a whim, and eventually, I became surrounded by new friends who were among some of the hardest working students in the country. By my senior year, I was ranked as a top ten debater by the National Speech and Debate Association. Debate required me to study topics such as normative ethics, epistemology, philosophy of science, and metaphysics which helped me find a new ambition for physics and math. Later in high school, I became fascinated with computers when I learned that programming came very naturally to me. Encouraged by finding this amazing field, I started taking school more seriously. I graduated high school as President of the National Honor Society, and with an associate degree from the Des Moines Area Community College.

I enrolled at Iowa State University to study Quantum Information Science. This was the perfect combination of my interests. The field allowed me to develop an understanding of the fundamental nature of the universe while also continuing to engage with topics that had far-reaching philosophical implications. I joined a quantum computing club freshman year, which was only 3 people: myself and two seniors. As a group, we were interested in quantum compression algorithms, but my understanding of the concepts was still limited at this point. To develop more skills, I would first complete multiple interdisciplinary research projects before later returning to quantum computers.

Since the beginning of college, I have been training to handle intense workloads by strengthening my ability to concentrate for long periods of time. Although this may come naturally for others, it took significant effort on my part to perfect my study habits. As a demonstration of my effort, this semester I have a 17 credit course load, a 10 hour per week teaching assistant position, a 40 hour per week research position at Ames National Lab, and I am the president of Iowa State's chapter of the IEEE Eta Kappa Nu honor society. Remaining driven and studious during the long semesters is incredibly difficult at times, but I believe I have continued to maintain this level of rigor because of my determination of being a competitive graduate applicant. An appropriate quote from Nietzsche that I like to remind myself of, "He who has a way to live can bear almost any how." This speaks to how a passion for science can act as my primary drive in school. Each semester I have performed well in my classes while simultaneously holding a teaching assistant job and conducting a research project. Being able to balance this type of schedule is exactly suited for graduate school.

My research experience has also granted me a perspective on the type of thinking that is necessary for scientific advancement. I firmly believe that the most critical aspect of the scientific process begins with an ability to ask big questions. Being confident and purposeful about your curiosity is necessary to make big scientific discoveries. My favorite example is Sir Roger Penrose's famous philosophy called Orchestrated Objective Reduction. Penrose's theory about consciousness emerging from quantum mechanical phenomena is famously criticized by other scholars; and yet, Penrose remained respected to such a degree that he was awarded a Nobel Prize. The biggest success I have had as an undergraduate came from partaking in the research process while being confident enough to let my own ideas guide each project. From my academic history, debate background, and scientific perspective I bring a unique ability to interpret, communicate, and initiate research.

The way I would like to impact others and advance society most directly is by becoming an educator. My first teaching experience was as an instructor at the National Symposium for Debate, an intense academic camp for high schoolers preparing to compete on the national debate circuit. I have also been a teaching assistant at Iowa State for the last 6 semesters. I continued holding teaching roles no matter how busy my schedule was because I greatly enjoy working with students. Furthermore, as President of Eta Kappa Nu, I am responsible for running the peer-tutoring help room, skill-building workshops, and problem-solving competitions we created for the benefit of students in the department. I have been a coach for high school debate, a tutor for computer science, and a physics outreach member. As a part of Eta Kappa Nu, Tau Beta Pi, and Phi Kappa Beta honor societies I participated in a variety of volunteer events both on campus and in our community. I have additionally been a part of the Engineering Student Council, and my department's Student Advisory Council as an undergraduate leader, establishing myself as a communicator between students and faculty.

One of the most important lessons I learned was the reason why producing knowledge is valuable. I learned that any academic venture is only important if you can express such found knowledge to the next generation to come. Debate had taught me this; that no matter how good your argument and ideas are, you only win rounds by getting the judge to understand you. I want to embrace this idea myself as a future teacher. Inspired by content creators like Grant Sanderson, Lex Fridman, and Andrew Dotson, I dream of about using graduate school as an opportunity to start a platform for my own content creation, with an emphasis on accessibility and explanation of profound scientific topics to a general audience.

More than just teaching, the way I would like to more positively impact students is by becoming involved with the administration of the department and organization of the curriculum. As a student I learned to appreciate the thrill of learning: thinking about the biggest questions of the universe and engaging in complex thought about the nature of reality and ourselves. Consequently, I would like to impact the following generations as best I can unearth each person's own academic prowess, like how I once miraculously found my own motivation. As a researcher, I am unsure if I will ever come close to standing next to the greatest scientists I admire. Donald Knuth described the type of person with a mind capable of such advanced problem solving as a "geek" and I accept that maybe I will never be able consider myself one. Nevertheless, I am deeply proud of the undergraduate career I have had and can confidently say I put forth my best and strongest effort. I received great grades with a challenging course load, all while balancing teaching, research, and involvement with the university. I believe I am ready to take everything I have learned thus far and continue my work at the next stage of my education.